

Summary

1.06.06

Human Interface Four Laws

1. A computer shall not harm your work or, through inaction, allow your work to come to harm
2. A computer shall not waste your time
3. An interface is humane if it is responsive to human needs and considerate of human frailties
4. The user should set the pace of interaction

Sampling of Design Methodologies

- Human-centered Design
- User-centered Design
- Usage-centered Design

1.07.04 UI

User Centered Design

- Use people and user profiles
- Scenarios / story narratives
- Too much focus on users may be harmful

Usage-Centered Design

- Think as the cases
- Model-driven process
- Focus on usage, not users
- Models of roles, tasks, and content

We need a Hybrid user/usage centered

Analysis: understand users, goals, skills, scenarios, requirements

Design: based on previous analysis

- Design doesn't start with pictures
- Starts with understanding people
- Software is means to an end

Understand:

- General reasons for Using Software
- Keys

- Ask right questions
- Solve the right problem
- Research users
- Ask why? 5 times

User Research Basics What to learn:

- Different user classes
- User goals
- Specific tasks
- Language they use
- Software skills
- Attitudes

Methods

- Direct observation
- Interviews
- Case studies
- Surveys
- Personas

Interface idioms

Examples:

- Form-based applications
- Text editors
- Graphic editors
- Spreadsheets
- Browsers
- Media Players
- Immersive games
- Web pages
- Social spaces
- E-commerce Sites

2.07.02

Determining Content

Capture/refine actors n goals

Analysis objects and relationships

Define/refine requirements

Explore scenarios

Use: use cases, class diagrams, state charts

List of Stuff

- Organizing collections is inevitable
- Types Objects, actions/tasks, tools
- Organize by category

Physical Window Structure

- Multiple windows
- One window paging
- Tiled panes Evaluate benefits, limitations, trade-offs

Design Patterns

Common solution to recurring problem

Solution: what n how

Problem: when

Context: why

Purposes

Show one single item
Showing list of things
Providing tools to create
Facilitating single task

Patterns

Feature Search & Browse
News stream
Picture Manager
Dash board
Settings Editor
Many Workspaces

Physical Structure Patterns

- Two panel selector
- Canvas plus palette
- One-window drilldown
- Alternative view
- Wizard
- Extras on demand
- Intriguing Branches
- Multi-level help

2.01.03

Key Concepts

• Elements

- Visual hierarchy
- Visual flow
- Grouping/alignment
- Dynamic displays
- Integration

• Hierarchy mechanisms

- Upper-left preference, whitespace
- Contrasting fonts/colors
- Positioning

• Gestalt principles

- Proximity
- Similarity
- Continuity
- Visual closure

• Dynamic features

- Scroll regions
- Collapsible/movable panels
- Card stacks

• Layout patterns:

- Visual Framework, Center stage, Titled Sections
- Responsive Disclosure/Enabling, Liquid Layout, Grid of Equals, Accordion

Key Concepts

Goal: Present data visually to impart knowledge using graphics, tables, trees, maps

Organizational models: linear, tabular, hierarchical, network, geographic.

Perceptive variables: Color (hue/brightness/saturation), texture, position, orientation, size, shape

Navigation: Scroll/pan, zoom, drill, down, magnify

Data manipulation: Sort (alphabetically, numerically, by date/category), filter, query

Meaning: Labels, legends, axes, datatips, data brushing